Lemma The sectional curvature function $K(\cdot,\cdot)$ determines $Rm(\cdot,\cdot,\cdot,\cdot)$

Proof: $K(X,Y)$ determines $Rm(X,Y,Y,X)$

Using the symmetries, we can show this determines $Rm(\ \ )$ completely

$=$

Interpreting Ricci in terms of sectional:

Recall that $Ric(Y,Z) = \sum_{i=1}^{n} Rm(E_i, Y, Z, E_i)$ for $\{E_i\}$ orthonormal

(bilinear and symmetric $\Rightarrow$ determined by diagonal values)

Given a unit vector $V \in T_pM$, set $E_1 = V$ and complete to an orthonormal basis $E_2, \dots E_n$

Then $K(V, E_i) = Rm(E_i, V, V, E_i)$

$$\Rightarrow \quad Ric(V,V) = \sum_{i=2}^{n} K(V, E_i)$$

$\therefore Ric(V,V)$ = "average" sectional curvature over sections containing V

$=$

Special case: $\dim = 3$

$$\begin{pmatrix} Ric(E_1,E_1) \\ Ric(E_2,E_2) \\ Ric(E_3,E_3) \end{pmatrix} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} K(E_1,E_2) \\ K(E_2,E_3) \\ K(E_3,E_1) \end{pmatrix}$$

invertible $\Rightarrow$ Ric determines K (and Rm) for $\dim = 3$

$=$

Scalar curvature $S = \operatorname{tr} Ric$

$$\Rightarrow \quad S = \sum_{j \neq k} K(E_j, E_k) \quad \text{for an orthonormal basis}$$

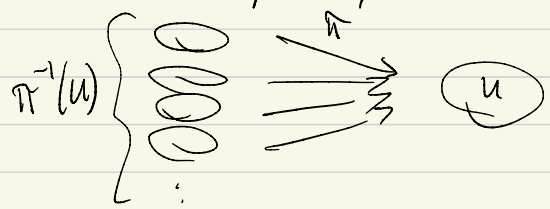
$\therefore S$ = "average" sectional curvature over all sections

Simple curvature theorems:

1) Cartan - Hadamard 1898, 1928

If M is a complete Riem mfld, with negative sectional curvature ($K(\cdot,\cdot) \leq 0$ at all pts) then for any $p \in M$, $\exp_p : T_pM \to M$ is a covering map.

covering map: $\pi : \tilde{M} \to M$ is a smooth surjective map such that every pt of M has a nbhd U such that each component of $\pi^{-1}(U)$ maps diffeomorphically to U



(CH $\to$ the "universal cover" of M is $\cong \mathbb{R}^n$)

2) Bonnet 1867 (surfaces)

Suppose M is a complete Riem. mfld. with
$K(\cdot,\cdot) \geq 1/R^2 > 0$

Then M is compact and $\operatorname{diam} M \leq \pi R$

(In fact the proof $\Rightarrow$ the universal cover of M is compact)

3) The classification of mflds with constant curvature

$K(\cdot,\cdot) =$ constant

By scaling we can assume $K(\cdot,\cdot) = -1, 0,$ or 1

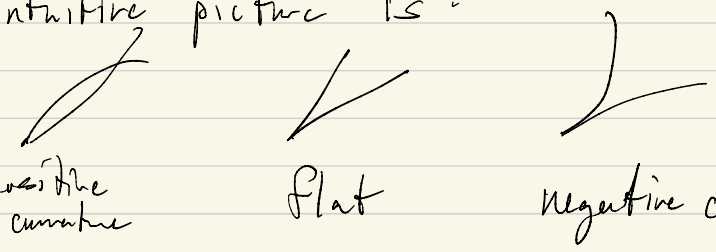
The universal cover is isometric to

$\begin{cases} S^n & K=1 \\ \mathbb{R}^n & K=0 \\ \mathbb{H}^n & K=-1 \end{cases}$

Jacobi fields

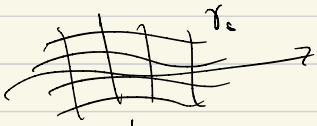
Our goal is to understand how geodesics behave under curvature assumptions

The intuitive picture is:

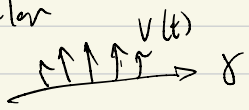


positive curvature | flat | negative curvature

Consider a geodesic $\gamma(t)$ extended to a variation $\Gamma(s,t)$ (smooth)



For each s the curve $\gamma_s = \Gamma(s,\cdot)$ is regular



The variation field of Γ is $V := \partial_s \Gamma|_{s=0}$

We defined $S := \partial_s \Gamma$, $T = \partial_t \Gamma$, so $V(t) = S(0,t)$

Symmetry Lemma: $D_t S = D_s T$

For the proof, we used the fact that $[S,T] = 0$

Cor For a vector field Z along Γ,

$$D_s D_t Z - D_t D_s Z = R(S,T)Z$$

Proof: By def.,

$$R(S,T)Z = \nabla_s \nabla_t Z - \nabla_t \nabla_s Z - \nabla_{[S,T]} Z$$

$[S,T] = 0$

Thm (Jacobi equation)
Suppose Γ is a variation of γ such that the curves γ_s are geodesic.
Then the variation field V satisfies

$$D_t^2 V + R(V, \dot\gamma)\dot\gamma = 0$$

Proof: γ_s geodesic means $D_t T = 0$ on Γ

Apply D_s: $\quad 0 = D_s D_t T$

$$= D_t D_s T + R(S,T)T \quad \text{by } [S,T]=0$$

$$0 = D_t^2 S + R(S,T)T \quad \text{symmetry}$$

Then use $S(0,t) = V(t)$, $T(0,t) = \dot\gamma(t)$

Def'n: A Jacobi field is a vector field V along a geodesic γ such that

$$D_t^2 V + R(V,\dot\gamma)\dot\gamma = 0$$

In local coord's: choose an orthonormal frame $\{E_i\}$ which is parallel along γ

$$\langle E_i, E_j\rangle = \delta_{ij}, \quad D_t E_j = 0$$

Let $V = v^i E_i$

parallel frame $\Rightarrow$ $D_t^2 V = \ddot v^i E_i$

The Jacobi eq. becomes: $\ddot v^i + R^i_{jkl} v^j \dot\gamma^k \dot\gamma^l = 0$

2nd order linear system $\Rightarrow$ existence + uniqueness
given $v^i(0)$ and $\dot v^i(0)$ $\quad i = 1, \dots, n$

Prop. Given a geodesic γ with $p = \gamma(0)$ and $X, Y \in T_pM$,
$\exists$ a unique Jacobi field J with

$$J(0) = X, \quad D_t J(0) = Y$$

trivial cases:

$J_0(t) = \dot{\gamma}(t)$ corresponds to $\Gamma(s,t) = \gamma(t+s)$

$J_1(t) = t\dot{\gamma}(t)$ " " $\Gamma(s,t) = \gamma(e^s t)$

Note that these are both tangential to γ

def: A <u>normal</u> Jacobi field along γ satisfies $J \perp \dot{\gamma}$ at all pts.